Cost of producing electric power

Plant manager given budget of $165M to produce 150K kilowatt hours at $1,100

	Cost	
Budgeted		
Unit cost	$1,100	
Q	150,000	
Cost	$165,000,000	
Actuals		
Unit cost	$1,125	
Q	160,000	
Cost	$180,000,000	
Total variance =	$15,000,000	U
Price variance = Q*(P - EP)	$4,000,000	U
Quantity variance = EP*(Q - EQ)	$11,000,000	U
Total variance =	$15,000,000	U

Mesla forecasts Revenue of $7B, based on selling 100K vehicles at $70,000 each.

		Model S	
Budget			
	EP	$70,000	
	EQ	100,000	
	Revenue	$7,000,000,000	
Actual			
	P	$67,000	
	Q	103,500	
	Revenue	$6,934,500,000	
	Total variance =	-$65,500,000	U
	Price variance = Q*(P - EP)	-$310,500,000	U
	Quantity variance = EP*(Q - EQ)	$245,000,000	F
	Total variance =	-$65,500,000	U

Mesla is budgeting to sell 100K Model S @ $70K and 200K Model 3 @ $38K.
It ends up selling a total of 304K vehicles (more than goal) at prices of $72K and $40K.

	Model S	Model 3	Total	
Budgeted				
EP	$70,000	$38,000	$48,667	
EQ	100,000	200,000	300,000	
Revenue	$7,000,000,000	$7,600,000,000	**$14,600,000,000**	
Actuals				
P	$72,000	$40,000	$47,789	
Q	74,000	230,000	304,000	
Revenue	$5,328,000,000	$9,200,000,000	**$14,528,000,000**	
Total variance =	-$1,672,000,000	$1,600,000,000	-$72,000,000	U
Price variance = Q*(P - EP)	$148,000,000	$460,000,000	$608,000,000	
Quantity variance = EP*(Q - EQ)	-$1,820,000,000	$1,140,000,000	-$680,000,000	
	-$1,672,000,000	$1,600,000,000		

We can think about this in terms of this variance decomposition or in terms of average selling price:
Ended up selling 304,000 units at $47,789 rather than 300,000 at $48,667.

The exact same analysis could be applied to margins -- gross margin could fall even as both price and volume exceed targets for the individual products.

The exact same analysis can be applied to costs of production.